ANALYTICS FOUNDATIONS · CRM SALES

Introduction to Data Analytics & Business Intelligence

Turn business data into decisions.

Understand analytics and BI, operational and analytical systems, data sources and formats, ETL and ELT, warehouses, lakes and lakehouses.

Applied throughout to CRM sales analytics.



A useful dashboard begins with a decision

- Are we achieving our revenue target?
- Which regions and products perform best?
- Where are opportunities getting stuck?
- Which customers contribute the most revenue?
- What needs attention this week?

01	02	03	04	05
Business question	**Required data**	**Analysis**	**Insight**	**Action**

Data analytics examines data to understand performance and find patterns

Data

Recorded facts.

Individual sales records.

→

Information

Organized facts.

North generated ₹18 lakh this month.

→

Insight

Interpret a pattern.

North missed target because two large deals slipped.

→

Action

Make an informed decision.

Review those deals and assign follow-up owners.

Different questions need different kinds of analytics

Descriptive

What happened?

Revenue fell by 10%.

Diagnostic

Why did it happen?

Fewer opportunities won in the South.

Predictive

What might happen?

Which open deals are likely to close?

Prescriptive

What should we do?

Which deals should the team prioritize?

BI makes trusted business information accessible and repeatable

Connect

Combine business data and define consistent metrics.

Explore

Present reports and dashboards; let users filter and investigate.

Share

Distribute information through controlled access.

One agreed definition of revenue across executive, regional and salesperson reports.

Reports, dashboards, metrics and KPIs each play a role

Report

Detailed analysis across views

Example: Monthly sales performance report

Dashboard

At-a-glance view of key information

Example: Executive sales overview

Metric

A numerical measurement

Example: Total revenue

KPI

A metric tied to an objective

Example: Revenue target achievement

₹40 lakh

Revenue

₹50 lakh

Target

80%

Achievement

Running the business and analyzing it serve different needs

Operational (OLTP)

Main purpose: Process business transactions

Typical work: Insert or update individual records

Data focus: Current business state

Design emphasis: Reliable, fast transactions

CRM example: Update an opportunity stage

Analytical (OLAP)

Main purpose: Explore performance and trends

Typical work: Read and aggregate many records

Data focus: Historical and integrated data

Design emphasis: Efficient analytical queries

CRM example: Compare regional win rates

CRM data needs an analytical layer for history, context and consistency

CRM captures change

Records change as the sales process progresses.

Today's opportunity stage cannot reveal time spent in each earlier stage.

Analytics adds history

Retain historical snapshots.

Analytics adds context

- Combine finance, targets and product data
- Simplify reporting with business-friendly models
- Share calculations for consistent results

Data format affects how we store and process information

Structured

Characteristics: Defined columns and data types

CRM example: Customer and sales tables

Semi-structured

Characteristics: Keys or tags, flexible fields

CRM example: JSON returned by a CRM API

Unstructured

Characteristics: No fixed tabular schema

CRM example: Emails, documents, call recordings

Business insights usually need more than one system

CRM

Leads, customers, opportunities, activities

Analytical purpose: Pipeline and conversion

ERP / Finance

Orders, invoices, payments

Analytical purpose: Financial and sales outcomes

Product systems

Products, categories, prices

Analytical purpose: Product performance

Excel / CSV

Targets, budgets, mappings

Analytical purpose: Actual vs target

Applications / APIs

Usage and interaction events

Analytical purpose: Engagement analysis

Connect through databases, files, APIs, events or approved exports.

ETL transforms data before loading

Extract

Read data from source systems.

Extract sales files.

→

Transform

Clean, standardize and reshape.

Standardize regions and dates.

→

Load

Write prepared data into the destination.

Load reporting tables.

Useful when data must be validated, reduced or protected before entering the destination.

ELT loads data before transforming

Extract

Read source systems.

→

Load

Place CRM extracts in an approved raw landing area.

→

Transform

Clean customer records using the destination platform.

→

Reuse

Create analytical tables and revenue summaries.

Useful when the platform scales processing and retains source data under control.

Choose the transformation location to suit the requirement

ETL

Transformation occurs: Before loading the target

Processing location: Integration or staging environment

Raw source retention: Depends on the design

Common strength: Prepare data before target entry

CRM example: Clean before warehouse load

ELT

Transformation occurs: After loading the target

Processing location: Destination platform

Raw source retention: Often kept in a landing layer

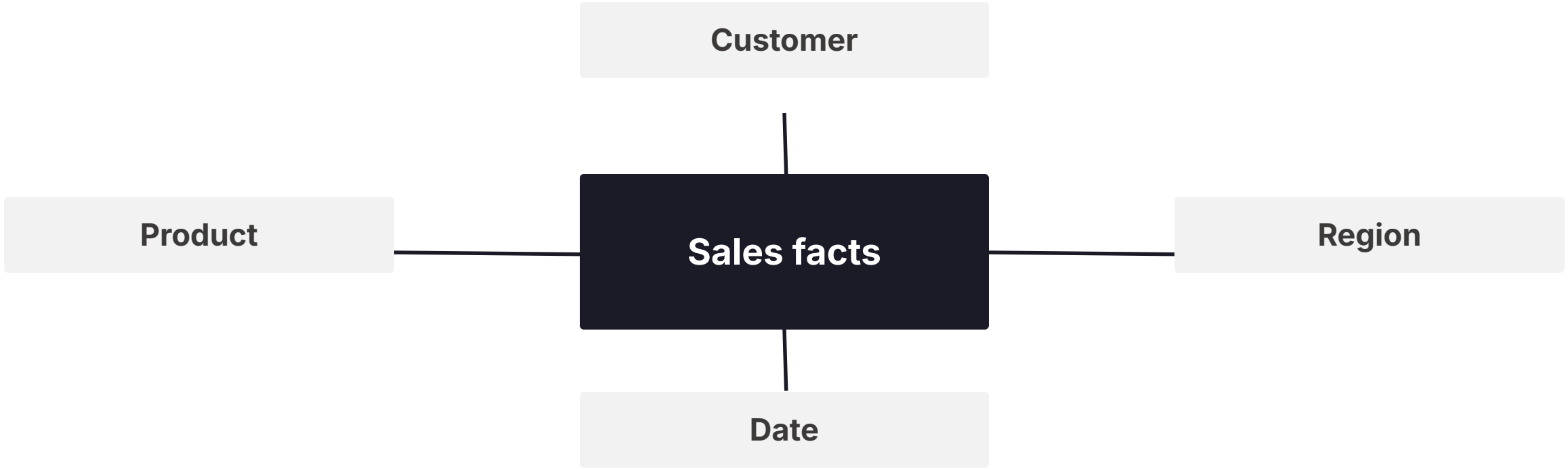
Common strength: Reprocess data within the platform

CRM example: Load raw, then build curated tables

A single solution can combine both approaches.

A data warehouse organizes curated data for consistent reporting

Integrated analytical data supports SQL queries, aggregations, shared definitions and recurring reporting.



A simple star schema links sales facts to descriptive dimensions.

A data lake offers flexible storage for many formats

Structured

CSV sales extracts.

Semi-structured

JSON opportunity records.

Unstructured

Customer documents.

Retain source files for future processing, exploration, engineering and data science.

Cataloging, access controls and lifecycle management are essential. Curated layers enable trusted consumption.

A lakehouse combines lake storage with managed analytical tables

Lake storage

Preserve raw CRM extracts and other files.

+

Managed analytical tables

Build curated Delta tables with reliable data management.

+

Shared consumption

SQL and other tools support engineering and analytical workloads.

Modeling, quality controls and governance remain essential.

Warehouse, lake and lakehouse have different strengths

Warehouse

Primary emphasis: Curated analytical data

Typical content: Structured business tables

Common use: BI and SQL reporting

CRM example: Sales reporting model

Data lake

Primary emphasis: Flexible data storage

Typical content: Files in many formats

Common use: Raw retention and exploration

CRM example: Original CRM exports

Lakehouse

Primary emphasis: Lake storage with managed tables

Typical content: Files and analytical tables

Common use: Engineering, analytics, BI

CRM example: Raw-to-curated CRM platform

These architectures can coexist in one enterprise solution.

A modern analytics platform runs from source data to business action

01

Ingest

Collect approved sources.

02

Store

Retain files and tables.

03

Transform

Clean and integrate.

04

Model

Define relationships and measures.

05

Consume

Explore reports.

06

Act

Guide business decisions.

Across every stage: security, governance, monitoring and ownership.

Bronze, Silver and Gold layers progressively refine data

Bronze

Preserve ingested source data with traceability

CRM example: Original customer and sales extracts

→

Silver

Validate, standardize, integrate

CRM example: Clean customers, consistent region codes

→

Gold

Prepare business-ready analytical data

CRM example: Sales facts, dimensions, KPI summaries

Each layer needs clear quality expectations and ownership.

COURSE USE CASE

Use case: a CRM Sales & Revenue Analytics Platform

Performance

- Track revenue against targets
- Compare salesperson and regional performance
- Identify valuable customers

Pipeline

- Understand the opportunity pipeline
- Find opportunities needing follow-up

Core datasets: Customers, Leads, Opportunities, Sales, Products, Salespeople, Targets, Activities.

First ask: what does one row represent?

Customers

One row per customer

Key relationship: CustomerID

Opportunities

One row per opportunity

Key relationship: CustomerID, SalespersonID

Sales

One row per sales line

Key relationship: CustomerID, ProductID

Products

One row per product

Key relationship: ProductID

Targets

One row per salesperson per month

Key relationship: SalespersonID, Month

Check uniqueness and relationship cardinality before combining tables.

Dimensions describe, measures count, KPIs assess

Dimension

By whom, where, what, when?

CRM examples: Salesperson, Region, Product, Date

Measure

How much or how many?

CRM examples: Revenue, Quantity, Opportunity Count

KPI

How well are we meeting an objective?

CRM examples: Target Achievement %, Win Rate

Analyze Revenue by Region and compare Target Achievement %.

A KPI formula needs a clear business rule

Total Revenue

Sum of valid sales revenue for the period

Pipeline Value

Sum of estimated values of open opportunities

Win Rate

$$\text{Won} \div (\text{Won} + \text{Lost}) \times 100$$

Average Won Deal Size

$$\text{Total won value} \div \text{won count}$$

Target Achievement

$$\text{Actual revenue} \div \text{target} \times 100$$

Lead Conversion Rate

$$\text{Converted} \div \text{eligible leads} \times 100$$

Define time period, eligible records, status rules, currency and zero-denominator behavior.

Data quality: can we trust the result?

Issue	Example	Potential impact
Duplicate records	Customer appears twice	Inflated customer counts
Missing values	Region is blank	Incomplete regional analysis
Inconsistent labels	North, NORTH, N. Region	Fragmented totals
Invalid values	Close date before creation date	Incorrect sales-cycle analysis
Broken references	Sale uses unknown ProductID	Missing product attribution
Wrong grain	Order total repeated on every line	Overstated revenue

PRACTICAL EXERCISE

Hands-on 1: explore CRM business data

1. Inspect the supplied CRM datasets
2. Identify the grain of each table
3. Find identifiers and relationships
4. Classify dimensions and measures
5. Define five metrics or KPIs
6. Record data-quality issues
7. Propose three dashboard questions

30–40

minutes

Deliver a short CRM data-discovery worksheet.

Document your findings

Dataset / field	Classification or grain	Validation question
Customers	One row per customer	Is CustomerID unique and populated?
Sales.Revenue	Measure	Line revenue or repeated order total?
Customers.Region	Dimension	Are region names consistent?
Opportunities.CustomerID	Relationship field	Does the customer exist?
Targets	Salesperson-month grain	Is each target recorded once?
Win Rate	KPI	Which statuses and dates are included?

Complete when you can explain each dataset, its relationships, supported questions, quality issues to investigate and KPI definitions needing agreement.